Tianhao Li

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Education

North China University of Technology, Beijing, China **September 2020 – July 2024 (Expected)**
Undergraduate student in the branch of Information Security *GPA: 3.88/4 Grade: 90.69/100 transcript*

Core Professional Courses: Applied Cryptography, Data Security, Image Processing, Principles of Computer Composition, Operating System, Reverse Engineering, Digital Forensic, Software Security, Network Attack and Defense

Skills

Programming Language: C/ModernC++(3 years+, expert), Python(3 years+, expert), Rust(1 year+, experienced), Java(1 year+, intermediate), Assembly(1 year+, experienced), Matlab(1 year+, intermediate), Shell(3 years+, intermediate)
OS & Tools: GNU/Linux(CLI&GUI, 4 years+, expert), Microsoft Windows(14 years+, expert), gcc, g++, gdb, make, cmake, vim, tmux, ssh, jupyter, etc.
Software Framework: PyTorch, PaddlePaddle, Tensorflow, TensorRT, Django, Vue.js, OpenSSL, GmSSL, etc.
Natural Language: Native Mandarin, Fluent Professional English (CET-4: 552, CET-6: 529, IELTS: 6.5)
Teamwork & Documentation: Git, SVN, LaTeX, Markdown, Github, Gitlab, TortoiseSVN, Overleaf, Zotero
Miscellaneous: Driving (Since Sep 2020), Photography, Adobe PS/PR/AE/AI/AU, Cardiopulmonary Resuscitation (CPR)

Experience

NSFOCUS Technologies Group Co.,Ltd **September 2023 – Now**
Security Research Intern @ Innovation Institute Dubhe Lab *Beijing, China*

- Duty: **a)** Completing the literature review work in the field of Trustworthy AI, and outputting technical reports related to the security risks of AI applications based on the latest scientific research results in the academic world. **b)** Participated in the design and development of an AI security detection platform of the Ministry of Industry and Information Technology (MIIT), and deployed an demo of advhat face recognition attack algorithm based on Python3 and Tensorflow framework. **c)** Participate in AI-related attack and defense competitions (e.g., DataCon, etc.) to analyze and leverage AI's own security competition datasets.
- Involved Techs & Skills: Academic English, Literature Reading, Python, PyTorch, Tensorflow, GNU/Linux, Adversarial Attacks, Backdoor Attacks, Deep Neural Networks, Machine Learning.

China Telecommunications Group Co.,Ltd **July 2023 – September 2023**
Data Analysis Intern @ Integration And Software Development Center *Suzhou, China*

- Duty: **a)** Utilizing Python to clean, organize, and preprocess large volme of complex data in order to facilitate subsequent analytical tasks. **b)** Collaborated with colleagues to investigate and understand user's requirements, thereby adjusting data analysis methods and objectives to ensure that the analytical outcomes meet users' practical needs.
- Involved Techs & Skills: Python3, Data Cleaning, Data Analysis, Efficient Communication, Agile Process

High-Tech Innovation Institute **July 2022 – July 2023**
Research Intern @ VR & AR Technology Innovation Center *Beijing, China*

- Duty: **a)** Research and development on lightweight metahuman generation scheme based on 3D reconstruction algorithm and Alicevision framework. **b)** Research on classification of human emotion based on electroencephalogram (EEG) and Swin-Transformer. **c)** Research on early warning mechanism for aviation safety based on QAR stream data.
- Involved Techs & Skills: ML/DL, Data Mining, Python, PyTorch, Academic Writing, LaTeX, Scientific Drawing
- Mentors: Prof. Dr. Lijun Wang (RAE Fellow), Prof. Dr. Zhengping Li, Dr. Ying Li

Chenchuang Technology Development Co.,Ltd **January 2022 – March 2022**
C++ Software Development Intern *Chengdu, China*

- Duty: Develop a WebAssembly-based real-time streaming video decoding and object detection system
- Involved Techs & Skills: WebAssembly, Emscripten, FFMpeg, OpenCV, HTML/CSS/Javascript, C++

Projects

Authentication And Physical Attack Warning System Based on Gait Recognition **May 2023 – August 2023**
Team Leader, Backend Developer Demo Video 1 | Demo Video 2 | About Page

- **As team leader**: **a)** Determine team division of labor based on team members' abilities. **b)** Design system architecture. **c)** Completion of technical documentation.
- **As backend developer**: **a)** Train and deploy deep learning models in server end. **b)** Implement network module, including protocol design and server-end network programming. **c)** Implement crypto module, including secure key distribution on public network channel and encryption/decryption using asymmetric crypto system and sm2/sm3/sm4 algorithms.

- **Involved Techs & Skills**: C++11, Python3, PyTorch, PaddlePaddle, ByteTrack, PaddleSeg, GaitBase, MySQL, Digital Envelop, Digital Signature, Asymmetric Crypto System, LaTeX
- **Status**: This project has been piloted on a floor of China Telecom's office building, and has been well received by leaders and colleagues, and is still in the process of being iterated.

RAM File System Based on Microsoft FAT32 Format **Nov 2022 – Jan 2023**

Full-stack Developer Git | About Page

- **As full-stack developer**: **a)** Read the specification for Microsoft's FAT32 file system for top-level software engineering, including layered structure, layered functionality. **b)** Based on modern C++, STL libraries and BoostC++ libraries to implement the algorithms and functions involved in abstracting the memory hard disk layer, block read/write layer, block management layer, file management layer, and application layer. **c)** Project executable build based on CMake, data visualization based on fmt and tabulate, unit testing based on Boost. **d)** Write simple software documentation
- **Involved Techs & Skills**: C++11, Boost C++, C++ STL, CMake, Operation System, File System

Academic Achievements

Publications (*Corresponding Author)

- **Tianhao Li**, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(Status: Undergoing peer-review by PeerJ Computer Science, JCR-Q2, IF=4.1)*
- Weizhi Ma, Ying Li*, **Tianhao Li**, Haowei Yang, Zhengping Li, Lijun Wang, Junyu Xuan. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *(Status: Undergoing peer-review by IEEE Transactions on Affective Computing, JCR-Q1, IF=11.9)*
- Weizhi Ma, Yujia Zheng, **Tianhao Li**, Zhengping Li, Ying Li, Lijun Wang*. Deep Learning in EEG-Based Emotion Recognition: Progress and Practical Implications. *(Status: Undergoing peer-review by PeerJ Computer Science, JCR-Q2, IF=4.1)*
- **Tianhao Li**, Yujia Zheng, Weizhi Ma, Guangshuo Wang, Zhengping Li, Lijun Wang*. Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems. *(Status: Submitted to Journal of the Society for Information Display, JCR-Q3, IF=2.3)*
- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Jiaxiang Zheng, Zhengping Li, Lijun Wang*. Unveiling Privacy Challenges: Big Data-Driven Digital Twins in SmartCity Applications. *(Status: Submitted to 2024 International Conference on Display Technology, EI-Compendex)*
- **Tianhao Li**, Yujia Zheng, Haoan Zhang, Weizhi Ma, and Ying Li. 2023. Research on Real-time Early Warning Mechanism of Aviation Safety Based on Finite State Machine Underlying in QAR Stream Data. In Proceedings of the 2023 7th International Conference on Big Data and Internet of Things (BDIOT '23). Association for Computing Machinery, New York, NY, USA, 216–222. https://doi.org/10.1145/3617695.3617719 *(EI-Compendex, Scopus)*
- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Ying Li, Zhengping Li, Lijun Wang*. Automated Pricing and Replenishment Decision for Vegetable Products Based on Hybrid Machine Learning Models. *(Status: Accepted by The 5th International Conference on Machine Learning and Intelligent Systems, EI-Compendex, Scopus)*

Invention Patents

- Multi-view Person Autonomous Location Method Based on Gait And Face Recognition. *(Status: Pending)*
- Physical Attack Automatic Early Warning Method Based on Gait Whitelist Mechanism. *(Status: Pending)*
- A Universal Method for Psychological State Classification Based on Multidimensional Extraction of EEG Signal Features. *(Status: Pending)*

Talks

- A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(English Poster Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- Automated Pricing and Replenishment Decision For Vegetable Products Based on Hybrid Machine Learning Models *(English Oral Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- A Comprehensive Review of Deep Learning in EEG-based Emotion Recognition: Classifications, Trends, and Practical Implications *(English Oral Presentation)* @ 2023 5th International Conference on Machine Learning and Intelligent System (MLIS'23), Macau SAR, China. [certificate]
- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism *(English Oral Presentation)* @ 2023 7th International Conference on Big Data and Internet of Things (BDIoT'23), Beijing, China. [certificate][slide]

Honors and Awards

National Contests

- (Nov.2023, National, *Top9.8%*) Second Prize in 11th Digital Media Technology and Creativity Contest [certificate]
- (Nov.2023, National, *Top12.4%*) Third Prize in 8th National Cryptography Contest [list][news]
- (Aug.2023, National, *Top13.1%*) Third Prize in 16th CISCN Security Project Contest [cert.][report][slide][video]
- (Apr.2023, National, *Top5.7%*) First Prize in 13th MathorCup Mathematical Modeling Challenge [cert.][paper]
- (Jun.2021, National, *Top15.3%*) Third prize in 12th Lanqiao Software Development and Algorithm Contest [cert.]

Non-National Contests

- (Sep.2023, Provincial) Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling [cert.]
- (Jun.2023, Regional) Third Prize in 16th CISCN AWDP (Attack with Defense Plus) Contest [certificate]
- (Apr.2023, Provincial) First Prize in 14th Lanqiao Software Development and Algorithm Contest [certificate]
- (Jun.2022, Regional) Third Prize in 15th CISCN CTF (Capture the Flag) Contest [certificate]
- (Apr.2021, Provincial) First Prize in 12th Lanqiao Software Development and Algorithm Contest [certificate]

Scholarships

- (2022-2023, *Top0.15%*) Received scholarships for outstanding students, academic, innovation, athletes, moral.
- (2021-2022, *Top2.91%*) Received scholarships for outstanding students, academic, innovation, athletes, moral.
- (2020-2021, *Top17.61%*) Received scholarships for academic, innovation.

Certifications

- Certificate of completion of Digital Talent Development Programme from *Google* [certificate]
- Certificate of completion of Artificial Intelligence course from *China Computer Federation (CCF)* [certificate]
- Position Competency Assessment Certificate of Industry and Information Technology Talents from *Ministry of Industry and Information Technology of the People's Republic of China* [certificate]

Open Source Activities

- I am contributor of garak(Nmap for LLM), OpenGait(A framework for gait recognition), All-in-One-Gait(A gait recognition system), and missing-semester-cn.github.io(The computer science missing semester).

Extracurricular Activities

- Mountain Biking: See POV videos on bilibili channel. Top 9 racer in 2023 GDL Thaiwoo downhill race.
- Road/Trail Running and Hiking: Finisher in 2023 Chongli 168 Ultra Trail(Biggest in Asia) DTC-100KM race.
- Photography and Off-road Driving: Keen on wildlife photography and exploring the unknown.
- Voluntary service: Nearly 400 hours were recorded by Beijing Volunteer Association (As of July 2023)

李天昊

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litianhao.life 领英 Orcid Github 知乎 知乎 B站

教育背景

北方工业大学, 中国，北京 **2020年9月 – 2024年7月（预计）**

本科在读 - 信息安全 绩点：*3.88/4* 成绩：*90.69/100* 成绩单

核心专业课程: 应用密码学, 数据安全, 数字图像处理, 计算机组成原理, 操作系统, 逆向工程, 数字取证, 软件安全, 网络攻防

专业技能

编程语言: C/现代C++(3年, 熟练), Python(5年, 熟练), Rust(1年, 使用过), Java(1年, 中等), Assembly(1年, 使用过), Matlab(1年, 中等), Shell(3 years+, 中等)
操作系统与工具: GNU/Linux(命令行与图形界面, 4年, 熟练), Microsoft Windows(14年, 熟练), gcc, g++, gdb, make, cmake, vim, tmux, ssh, jupyter, etc.
软件框架: PyTorch, PaddlePaddle, Tensorflow, TensorRT, Django, Vue.js, OpenSSL, GmSSL, etc.
自然语言: 普通话, 流利的专业英语 (大学英语四级：552，大学英语六级：529，雅思考试：6.5)
团队合作与文档撰写: Git, SVN, LaTeX, Markdown, Github, Gitlab, TortoiseSVN, Overleaf, Zotero
杂项: 驾驶(从2020年9月), 摄影, Adobe PS/PR/AE/AI/AU, 心肺复苏(CPR)

工作经历

绿盟科技集团股份有限公司 **2023年9月 – 至今**

安全研究员实习 @ 创新研究院天枢实验室 中国，北京

- 工作职责: **a)** 完成TrustworthyAI（可信AI）领域的文献综述工作，根据学术界最新科研成果，输出人工智能应用安全风险相关技术报告。**b)** 参与工信部某AI自身安全检测平台的设计与研发，部署基于Python3语言与Tensorflow框架的advhat人脸识别攻击算法实例。**c)** 参加AI相关攻防竞赛（如DataCon等），针对AI自身安全竞赛数据集展开分析与利用。
- 技术与技能: 学术英语，文献阅读，Python，PyTorch，Tensorflow，GNU/Linux，对抗攻击，后门攻击，深度神经网络，机器学习。

中国电信股份有限公司 **2023年7月 – 2023年9月**

数据分析实习 @ 集成与软研中心 中国，苏州

- 工作职责: **a)** 利用 Python 对大量复杂数据进行清理、组织和预处理，以便于后续的分析任务。 **b)** 与同事合作，调查和了解用户的要求，从而调整数据分析方法和目标，确保分析结果符合用户的实际需要。
- 技术与技能: Python3, 数据清洗, 数据分析, 高效沟通, 敏捷过程

高精尖创新研究院 **2022年7月 – 2023年7月**

科研实习 @ 虚拟与增强现实技术创新中心 中国，北京

- 工作职责: **a)** 研发基于三维重建算法和Alicevision框架的轻量级数字人生成方案。 **b)** 基于脑电图和Swin Transformer的人类情感分类研究。 **c)** 基于QAR流数据的航空安全预警机制研究。
- 技术与技能: 机器学习与深度学习方法, 数据挖掘技术, Python3, PyTorch, 学术写作, LaTeX, 科研绘图
- 导师: 王立军院士, 李争平教授, 李颖博士

辰创科技发展有限公司 **2022年1月 – 2022年3月**

C++软件开发实习 中国, 成都

- 工作职责: 开发基于WebAssembly的实时流媒体视频解码和目标检测系统
- 技术与技能: WebAssembly, Emscripten, FFMpeg, OpenCV, HTML/CSS/Javascript, C++

项目实践

基于步态识别的身份验证和近源渗透预警系统 **2023年5月 – 2023年8月**

团队领导者, 后端开发者 演示视频1 | 演示视频2 | 介绍页

- **作为团队领导者**: **a)** 根据团队成员的能力确定团队分工。 **b)** 设计系统技术架构。**c)** 完成技术文档的撰写。
- **作为后端开发者**: **a)** 在服务器端训练和部署深度学习模型。 **b)** 实现网络模块，包括协议设计和服务器端网络编程。 **c)** 实现加密模块，包括在公共网络通道上安全分发密钥，以及使用非对称加密体制和sm2/sm3/sm4算法进行加密与解密。
- **技术与技能**: C++11, Python3, PyTorch, PaddlePaddle, ByteTrack, PaddleSeg, GaitBase, MySQL, 数字信封技术, 数字签名技术, 非对称密码体制, LaTeX
- **项目状态**: 本项目曾在中国电信办公楼某层进行试运行，获得了领导和同事的好评，仍在不断迭代中。

基于微软FAT32文件系统格式的内存文件系统的设计与实现 **2022年11月 – 2023年1月**

全栈开发者 Git | 介绍页

- **作为全栈开发者**: **a)** 阅读微软FAT32文件系统的技术手册，进行软件工程顶层设计，包括分层结构、分层功能。 **b)** 基于现代C++、STL库以及BoostC++库实现抽象内存硬盘层、块读写层、块管理层、文件管理层、应用层涉及到的各算法与函数。 **c)** 基于CMake进行项目可执行文件构建，基于fmt与tabulate进行数据可视化，基于Boost.Test进行单元测试。 **d)** 撰写简单的软件文档。
- **技术与技能**: C++11, Boost C++, C++ STL, CMake, 操作系统, 文件系统

学术成果

文章发表 (*通讯作者)

- **Tianhao Li**, Weizhi Ma, Yujia Zheng, Xinchao Fan, Guangcan Yang, Ying Li, Lijun Wang, Zhengping Li*. A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(Status: Undergoing peer-review by PeerJ Computer Science, JCR-Q2, IF=4.1)*
- Weizhi Ma, Ying Li*, **Tianhao Li**, Haowei Yang, Zhengping Li, Lijun Wang, Junyu Xuan. SFSWTS: EEG Emotion Recognition Based on Spatial-frequency Shifted Windows Time Self-attention Neural Network. *(Status: Undergoing peer-review by IEEE Transactions on Affective Computing, JCR-Q1, IF=11.9)*
- Weizhi Ma, Yujia Zheng, **Tianhao Li**, Zhengping Li, Ying Li, Lijun Wang*. Deep Learning in EEG-Based Emotion Recognition: Progress and Practical Implications. *(Status: Undergoing peer-review by PeerJ Computer Science, JCR-Q2, IF=4.1)*
- **Tianhao Li**, Yujia Zheng, Weizhi Ma, Guangshuo Wang, Zhengping Li, Lijun Wang*. Trustworthy Metaverse: A Comprehensive Investigation into Security Risks and Privacy Issues in Artificial Intelligence-Extended Reality Systems. *(Status: Submitted to Journal of the Society for Information Display, JCR-Q3, IF=2.3)*
- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Jiaxiang Zheng, Zhengping Li, Lijun Wang*. Unveiling Privacy Challenges: Big Data-Driven Digital Twins in SmartCity Applications. *(Status: Submitted to 2024 International Conference on Display Technology, EI-Compendex)*
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- Yujia Zheng, **Tianhao Li**, Weizhi Ma, Ying Li, Zhengping Li, Lijun Wang*. Automated Pricing and Replenishment Decision for Vegetable Products Based on Hybrid Machine Learning Models. *(Status: Accepted by The 5th International Conference on Machine Learning and Intelligent Systems, EI-Compendex, Scopus)*

发明专利

- Multi-view Person Autonomous Location Method Based on Gait And Face Recognition. *(Status: Pending)*
- Physical Attack Automatic Early Warning Method Based on Gait Whitelist Mechanism. *(Status: Pending)*
- A Universal Method for Psychological State Classification Based on Multidimensional Extraction of EEG Signal Features. *(Status: Pending)*

公开演讲

- A Survey on Gait Recognition Against Occlusion: Taxonomy, Dataset and Methodology *(English Poster Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- Automated Pricing and Replenishment Decision For Vegetable Products Based on Hybrid Machine Learning Models *(English Oral Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- A Comprehensive Review of Deep Learning in EEG-based Emotion Recognition: Classifications, Trends, and Practical Implications *(English Oral Presentation)* @ 2023第五届机器学习与智能系统国际会议 (MLIS'23), 中国, 澳门特别行政区 [证书]
- Dive Into QAR Stream Data - A Real-time Early Warning Mechanism *(English Oral Presentation)* @ 2023第七届大数据与物联网国际会议 (BDIoT'23), 中国, 北京 [证书][幻灯片]

荣誉奖项

国家级竞赛

- (2023年11月, 国家级, *Top9.8%*) 第十一届全国大学生数字媒体科技作品及创意竞赛二等奖 [证书]
- (2023年11月, 国家级, *Top12.4%*) 第八届全国密码技术竞赛三等奖 [名单][新闻]
- (2023年8月, 国家级, *Top13.1%*) 第十六届全国大学生信息安全竞赛作品赛三等奖 [证书][报告][幻灯片][演示][新闻]
- (2023年4月, 国家级, *Top5.7%*) 第十三届MathorCup数学建模挑战赛一等奖 [证书][论文]
- (2021年6月, 国家级, *Top15.3%*) 第十二届蓝桥杯全国软件和信息技术专业人才大赛三等奖 [证书]

非国家级竞赛

- (2023年9月, 省部级) 全国大学生数学建模竞赛北京赛区甲组二等奖 [证书]

- (2023年6月, 区域级) 第十六届全国大学生信息安全竞赛实践赛(AWDP)三等奖 [证书]
- (2023年4月, 省部级) 第十四届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]
- (2022年6月, 区域级) 第十五届全国大学生信息安全竞赛实践赛(CTF)三等奖 [证书]
- (2021年4月, 省部级) 第十二届蓝桥杯全国软件和信息技术专业人才大赛一等奖 [证书]

奖学金

- (2022-2023学年, *Top0.15%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2021-2022学年, *Top2.91%*) 荣获优秀学生、学习、创新实践、体育、思想品德奖学金。
- (2020-2021学年, *Top17.61%*) 荣获学习、创新实践奖学金。

荣誉证书

- 数字人才培养计划谷歌高校师资人才培训项目结业证书 来自 谷歌(Google) [证书]
- CCF群星计划之人工智能科普培训结业证书 来自 中国计算机学会(CCF) [证书]
- 工业和信息化人才岗位能力评价证书 来自 中华人民共和国工业和信息化部 [证书]

开源活动

- 我是garak(Nmap for LLM), OpenGait(A framework for gait recognition), All-in-One-Gait(A gait recognition system), and missing-semester-cn.github.io(The computer science missing semester)的开源贡献者。

课外活动

- 山地自行车: 可以访问B站主页观看视频。2023GDL太舞速降赛排名前九。
- 跑步与徒步: 荣获2023崇礼168超级越野赛DTC百公里组完赛奖。
- 摄影与驾驶: 热衷于野生动物摄影和探索未知世界。
- 志愿服务: 志愿北京平台记录近400小时。